

Calibration Collapse Under the Homology Cliff: Distant-Stratum Precision of 0.068 and ECE of 0.294 in Frozen ESM-2 Retrieval

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Abstract

Beyond the F_1 cliff characterized in Paper 1, frozen ESM-2 [1] cosine k -NN retrieval on our 24,885-protein biosecurity test set exhibits a severe calibration collapse across strata. Expected Calibration Error (ECE [2], six-bin reliability scheme) rises $4.3\times$ from close-stratum 0.069 to distant-stratum 0.294. Distant-stratum positive-prediction precision drops to $3/44 = 0.068$: of 44 distant queries where the classifier predicts positive, only 3 are actually positive. In the highest-confidence bin ($[0.9, 1.0]$ of vote fraction), close-stratum empirical positive rate is 0.91 while distant-stratum empirical positive rate is 0.00 (3 predictions, 0 correct). The classifier is most confident where it is least accurate on this stratum. The distant stratum is a natural covariate-shift regime, and Ovadia et al. [3] previously documented that standard deep-classifier post-hoc calibration techniques lose their guarantees under such shift; we measure but do not test those rescues here. Deployment consequence: in our regime, every distant-stratum positive prediction requires human review regardless of vote count, and distant-stratum positive outputs are not safely usable as automated triggers. Temperature scaling and isotonic regression [2] as post-hoc rescues are deferred to a follow-up. Headline values are committed as `data/results_summaries/calibration_results.json` and reproducible via `code/analyses/run_calibration.py`.

Erratum (2026-06-15, v1.2). The committed `calibration_results.json` that fed this paper predated a panel-construction change and did not reproduce from the current, verified pipeline; it has been regenerated. Corrected close/moderate positive-prediction precision are **0.788** and **0.347** (previously 0.891 and 0.467); close/moderate stratum sizes are **22,198** and **1,533** (previously 22,090 and 1,641); and the reliability figure is now plotted from the script-computed reliability table rather than hand-entered diagram read-offs. The headline distant-stratum precision ($3/44 = 0.068$), all three ECE values (0.069 / 0.154 / 0.294), and every conclusion are unchanged. See `CHANGELOG.md`.

1 Setup

Analysis at t_{30} $R = 1000$ $k = 25$ cosine, seed 20260410 (seed chosen before calibration analysis, from the main pre-reg). Vote fraction: $\hat{p}(y = 1) = \text{pos_votes}/k$. Prediction: $\hat{y} = \mathbb{I}[\text{pos_votes} \cdot 2 \geq k]$. Reliability binning: 6 bins at 0.1, 0.3, 0.5, 0.7, 0.9, 1.0. $\text{ECE} = \sum_b (|B_b|/n) \cdot |\text{acc}(B_b) - \text{conf}(B_b)|$.

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2 Results

stratum	n	ECE	positive-prediction precision
close ($s_{\max} \geq 0.95$)	22,198	0.069	0.788
moderate ($[0.90, 0.95)$)	1,533	0.154	0.347
distant (< 0.90)	154	0.294	0.068

Positive-prediction precision on distant stratum: 3 true positives out of 44 positive predictions. False alarm rate $41/44 = 93.2\%$ among distant positive predictions.

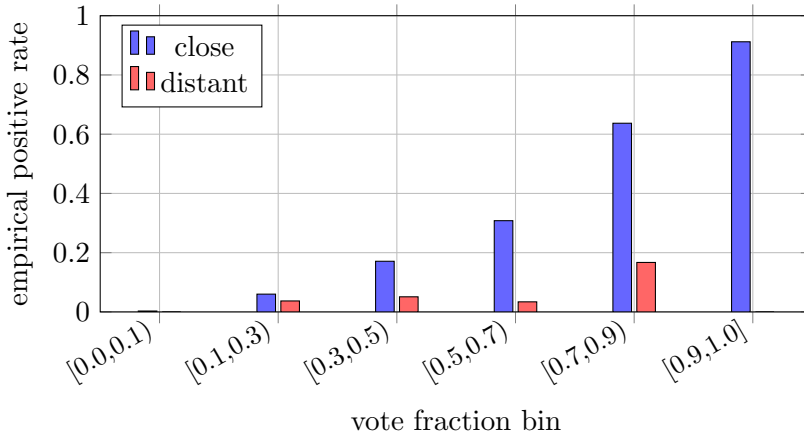


Figure 1: Close-stratum reliability rises with confidence and is well-calibrated in aggregate (ECE 0.069). Distant-stratum reliability is flat across confidence bins and collapses in the highest-confidence bin (3 predictions, 0 correct; ECE 0.294). Bars are the script-computed empirical positive rate per bin.

3 Why this matters

In biosecurity deployment, downstream filters often gate on vote-fraction confidence: route high-confidence positives to automated action, route low-confidence to review. Under the cliff, this policy fails catastrophically on the distant stratum: the single highest-confidence bin contains 3 predictions, all wrong, while adjacent lower-confidence bins have higher empirical rates. Confidence ranking within the distant stratum is essentially noise.

4 Related: calibration under distribution shift

Ovadia et al. [3] showed standard deep-classifier calibration methods (temperature scaling [2], deep ensembles) lose their guarantees under covariate shift. Our distant stratum is a natural covariate-shift regime induced by embedding-space distance from the panel. The standard correction family [2] does not straightforwardly apply because the shift is unbounded along the $s_{\max} < 0.90$ ray. Temperature scaling [2] and isotonic regression [2] are natural next-pass rescues, deferred. Panel augmentation is disfavored by the cross-family result in the companion Paper 5 (Wilson 95% CI on within-family rate $[0\%, 17\%]$); post-hoc distance substitution is rejected at pre-registered H1 by Paper 2 [1, 4–7].

5 Limitations

(L1) Single-seed analysis; 10-seed extension deferred. (L2) ECE is known to be sensitive to binning; we report a six-bin scheme with boundaries $\{0.0, 0.1, 0.3, 0.5, 0.7, 0.9, 1.0\}$ (tight tails, wide middle), per the locked binning in `code/analyses/run_calibration.py`. Re-binning may shift the absolute ECE values. (L3) Calibration under the learned-projection rescue (Paper 1) unmeasured. (L4) Isotonic regression and Platt scaling as post-hoc rescues untested.

References

- [1] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023.
- [2] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *ICML*, 2017.
- [3] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D Sculley, Sebastian Nowozin, Joshua Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model’s uncertainty? evaluating predictive uncertainty under dataset shift. In *NeurIPS*, 2019.
- [4] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.
- [5] Shun-ichi Amari. Information geometry and its applications. *Applied Mathematical Sciences*, 194, 2016.
- [6] Kilian Q Weinberger and Lawrence K Saul. Distance metric learning for large margin nearest neighbor classification. *JMLR*, 10:207–244, 2009.
- [7] Florian Schroff, Dmitry Kalenichenko, and James Philbin. FaceNet: A unified embedding for face recognition and clustering. In *CVPR*, pages 815–823, 2015.